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Cyber-physical customer management for internet of robotic things-enabled banking

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ABSTRACT In-person banking is still an important part of financial services around the world. Hybrid bank branches with service robots can improve efficiency and reduce operating costs. An efficient autonomous Know-Your-Customer (KYC) is required for hybrid banking. In this paper, an automated deep learning-based framework for interbank KYC in robot-based cyber-physical banking is proposed. A deep biometric architecture was used to model the customer's KYC and anonymise the collected visual data to ensure the customer's privacy. The symmetric-asymmetric encryption-decryption module in addition to the blockchain network was used for secure and decentralized transmission and validation of the biometric information. A high-capacity fragile watermarking algorithm based on the integer-to-integer discrete wavelet transform in combination with the Z_6 and A_6 lattice vector quantization for the secure transmission and storage of in-person banking documents is also proposed. The proposed framework was simulated and validated using a Pepper humanoid robot for the automated biometric-based collection of handwritten bank checks from customers adhering to COVID-19 pandemic safety guidelines. The biometric information of bank customers such as fingerprint and name is embedded as a watermark in the related bank documents using the proposed framework. The results show that the proposed security protection framework can embed more biometric data in bank documents in comparison with similar algorithms. Furthermore, the quality of the secured bank documents is 20% higher in comparison with other proposed algorithms. Also, the hierarchical visual information communication and storage module that anonymizes the identity of people in videos collected by robots can satisfy the privacy requirements of the banks. Overall, the proposed framework can provide a rapid, efficient, and cost-effective inter-bank solution for future in-person banking while adhering to the security requirements and banking regulations.

INDEX TERMS Cyber-physical banking, Internet of Robotic Things, Industry 4.0, interbank ecosystems, blockchain, deep learning

I. INTRODUCTION

During the COVID-19 pandemic, the banking sector delivered the majority of its financial services through online banking solutions. However, in-person banking services are still essential for the deposition and collection of handwritten bank checks and other traditional paper-based financial transactions. In addition, in-person banking services are useful for elderly customers who are unable to use digital banking. The major problems of in-person services in bank branches are their high cost, lack of seamless integration with digital banking, and lack of safety of interactions in pandemic conditions. Humanoid service robots acting as bank tellers and the Internet of Robotic Things (IoRT) [1] can provide a solution to these problems by creating hybrid

cyber-physical bank branches that are efficient, cost-effective, and safe.

Two important issues should be addressed in IoRT-based cyber-physical banking. The first is verifying the identity of the customers. The Know-Your-Customer (KYC) [2] guidelines as mandated by financial regulators are used in many financial institutions for the standard verification of a person requesting a financial transaction. The traditional KYC documents include identity documents, such as passports, driver's license, customer photographs, and signatures that are manually verified by the bank staff. However, traditional KYC processes are costly for banks and cumbersome for customers. Machine learning-based KYC verification can significantly enhance automation in banks, reduce costs, and increase the speed of the provision of

financial services. Unlike traditional KYC documents, such as passports, driver's licenses, and other identification documents that are vulnerable to fraud, the automated biometrics-based KYC is accurate, fast, and resilient to forgery.

The second issue is protecting the privacy of the customers. A major challenge to automated KYC solutions is ensuring the privacy and security of customer data. IoRT agents, acting as the edge nodes of the Internet of Things (IoT), lack the computational power to run the complex machine-learning models required for accurate biometric verification. It is thus necessary to transmit these data to the mainframe of the bank to receive the results of the validation. Another issue with traditional KYC is that the identification documents need to be separately verified by each bank, and customers are repeatedly subjected to initial and periodical KYC procedures. The secure transmission, processing, and storage of physical banking documents is another major issue in traditional in-person banking.

To address the above problems, this study proposes a blockchain-based framework that uses deep neural networks to validate a variety of biometrics-based information for KYC in IoRT-enabled cyber-physical banking. It is called the Automated Deep Decentralized KYC (ADD-KYC). In the proposed framework, humanoid robots, acting as Modular Rapidly Deployable Service Agents (MORAD-SA) [1], are used to provide in-person banking services and collect biometrics from customers. The blockchain network and smart contracts are used for the secure and decentralized transmission and validation of KYC information. Deep neural networks are used to validate the customers' biometrics. A KYC token allows the customer to have control over their KYC identity and, when necessary, share it with other banks. The proposed framework enables the ADD-KYC to be used across interbank ecosystems. Furthermore, a high-capacity watermarking algorithm is used to embed customer KYC data into in-person banking documents. A fragile high-capacity watermarking algorithm based on similar sub-lattices of the Z_6 and A_6 Lattice Vector Quantization (LVQs) is also proposed. The proposed algorithm is an improved version of the Z_4 and A_4 LVQ watermarking algorithms [4, 5]. It has a higher capacity as well as higher imperceptibility. The watermarking module ensures the secure transmission and archiving of bank documents.

The IoRT-based collection and validation of biometrics as well as various banking services are implemented using the humanoid robot Pepper, manufactured by Softbank Robotics. The automated collection and processing of handwritten bank checks from customers while adhering to safety guidelines followed during the COVID-19 pandemic are implemented as a case study. A complete scenario of the greeting of customers, their KYC verification, and the provision of various financial services using a service robot are applied. The problem of autonomous analysis and

information extraction from multi-language Persian–English bank checks is investigated using a handwriting and signature verification module.

In summary, to have an automated intelligent bank branch capable of providing traditional paper-based services to the customer using robots, a complex system capable of addressing multiple problems is required. The problems that the proposed ADD-KYC framework is addressing are as follows:

1. A practical robot-based secure hardware and software solution is required for banking. This solution reduces the cost of human resources and increases the speed and accuracy of the banking services as well as adhering to the safety requirements in pandemic conditions. The out-of-the-box humanoid robots require significant software modification to be able to provide specific banking services with high standards.
2. An automated multi-biometric KYC solution is required for secure banking. This system should allow the interbank ecosystem to provide customer identity verification with high accuracy.
3. A fast and high-capacity watermarking algorithm is required for securing digitally transmitted documents in the banking cloud. This algorithm should be capable of storing multiple biometric information in banking documents.
4. A decentralized solution is required for enabling the customers to have access to their digital KYC identity in the interbank environment while protecting the privacy and security of the customers.

To address the problems, the Add-KYC framework with the following contributions is proposed:

1. A blockchain-based decentralized framework is developed for the secure and privacy-preserving sharing of KYC information across the interbank ecosystem, and machine learning models are proposed to eliminate the need for KYC information collection across different banks.
2. An automated biometrics-based KYC solution is developed that is not dependent on identification documents that are vulnerable to fraud.
3. A fragile KYC watermarking solution for the secure transmission and archiving of in-person banking documents is proposed. It can embed a large amount of KYC data in bank documents with a high Peak Signal-to-Noise Ratio (PSNR).
4. A hierarchal privacy-preserving module is proposed that anonymises the identity of people in videos collected in the banking environment by IoRT agents for authorized-only access.
5. An IoRT-based hybrid banking solution that can act as the physical link between Banking as a Service (BaaS) and bank customers in normal as well as pandemic conditions is proposed.

Table 1 provides the abbreviations used in the manuscript and their corresponding descriptions.

TABLE 1. Abbreviations used in the manuscript and their corresponding descriptions.

Abbreviation	Description
KYC	Know-Your-Customer
IoRT	Internet of Robotic Things
ADD-KYC	Automated Deep Decentralized KYC
MORAD-SA	Modular Rapidly Deployable Service Agents
LVQ	Lattice Vector Quantization
PSNR	Peak Signal-to-Noise Ratio
BaaS	Banking as a Service
DWT	Discrete Wavelet Transform
BEMD	Bi-dimensional Empirical-mode Decomposition
DCT	Discrete Cosine Transform
SO	Particle Swarm Optimization
SVD	Singular Value Decomposition
FIWARE	Future-Internet-WARE
IPFS	Interplanetary File System
EdDSA	Edwards-curve Digital Signature Algorithm
RSA	Rivest–Shamir–Adleman
SHA	Secure Hash Function
AES	Advance Encryption Standard
ILDWT	Integer-Lifting DWT
CI	Content Identifiers
ReLU	Rectified Linear Unit
NDA	Non-Disclosure Agreements
LBPH	Locally Binary Pattern Histogram
CNN	Convolutional Neural Network

The remainder of this paper is organized as follows: Section II provides a review of the related literature and shortcomings in current methods. Section III details the proposed ADD-KYC framework, and Section IV discusses the implementation of IoRT-based service provision in banks and the results of experiments to verify the proposed framework. Section V summarizes the conclusions of this paper.

II. Background and related work

The KYC check is the mandatory procedure for verifying the identity of bank clients. It is among the most important procedures in banking systems and is periodically required by financial regulators in various countries for all financial transactions. The classic KYC procedure is the provision of identification documents and the address of residence of the bank client as well as a recent photograph and a sample of their signature. The biometrics-based KYC check has also been implemented in some banks around the world [2]. The classic method is the manual collection and verification of KYC data by the bank staff. However, service robots have recently proved to be more efficient and cheaper for the provision of bank services [3].

A major issue in the context of KYC checks is to attain a balance between accuracy and intrusiveness. Many forms of identity verification are time-consuming and lead to annoyance for customers. The applications of various biometrics, such as face [4], voice [5], iris [6], fingerprints, finger veins, palm veins, handwriting, and signature, have shown promising results for automated KYC. Facial and vocal biometrics are non-invasive and thus are more suitable for banking services among these. The advantages of deep neural networks for biometric feature extraction have been verified in the past few years [7]. Szczuko et al. [8] proposed a data fusion-based approach to multi-modal biometrics for the verification of bank clients. The Dempster–Shafer method was

used for data fusion, and the outcome was acceptably accurate for banking applications. Almadby and Elrefaei [7] reviewed recent advances in deep learning systems for applications of biometrics. Several performance indicators have shown that deep neural networks can provide the high accuracy required for verifying human identity. Estrela et al. [9] presented a behavioral measure of biometrics for user verification in banking applications to prevent fraud and impersonation attacks. It achieved accuracies of 90.68% and 97.05% when using biometrics for banking authorization in various scenarios. Hassan et al. [10] reviewed various soft and non-intrusive biometric measures and showed that they can provide accuracy comparable to that of classic biometric measures in various applications.

Another issue in automated biometrics is the security of the data and the privacy of the bank customers. The automated machine learning-based verification of biometrics requires the transmission of visual and other sensory feedback from edge nodes, such as robots, to the bank servers. Another issue is that each bank needs to separately train its biometric models, which intensifies the problems of customer irritation, data security, customer privacy, and KYC cost. To address these problems, blockchain-based KYC has been proposed. Laborde et al. [11] proposed a KYC system for online banking that combines identity attributes issued by different organizations. Jain et al. [12] proposed a decentralized framework for one-time KYC for easing interbank transactions. Biradar et al. [13] used Hyperledger Fabric, for a blockchain-based KYC framework to this end, and Yadav et al. [14] developed an Ethereum-based KYC model for improving customer experience and reducing costs. In this system, the bank customers are categorized into two groups: temporary clients requesting basic services, and permanent clients. Esposito et al. [15] proposed a blockchain-based framework for identity management and authorization that was integrated with the Future-Internet-WARE (FIWARE) platform. As applications of the smart city should be integrated with currently established ICT infrastructures within the city, the problem of data security is important. The centralized approach is not capable of addressing the requirements for security in such platforms and, therefore, a blockchain-based solution is proposed.

Table 2 shows research similar to the method proposed in this paper, but each of the relevant studies has a limitation that we address in the proposed ADD-KYC framework. Currently, available frameworks implement only the classic KYC on the blockchain while ADD-KYC is automated, based on a deep neural network, includes multiple biometrics, is designed for robotic banking, and features hierarchal privacy preservation. Furthermore, current methods for KYC validation used in the banking sector are manual, centralized, and document-based, and thus are time-consuming, costly, and not secure. By contrast, the proposed ADD-KYC framework is fast, non-intrusive, and secure.

An important issue regarding KYC is that banks are required to have strong security measures in place to prevent the unauthorized disclosure and use of the customer's KYC information. Therefore, the storage and transfer of the customer's KYC must be secure while being accessible by various interbank systems. The bank customers should have full control over their KYC and should be able to share it with other banks of their own will. In IoRT-based banking, a variety of biometrics-related information can be collected from the customers by the edge nodes. However, the customer should give the final permission for the KYC to be used.

Another important issue related to in-person banking is the security of physical documents that are scanned and transferred digitally across the banking ecosystem. The current solution for enhancing the security of these documents is encryption based on classic methods, such as AES. However, watermarking these documents using KYC information protects them from being tampered with, and this increases security. Various digital watermarking algorithms have been recently proposed [16]. Watermarking techniques can be divided into fragile and robust watermarking. Robust watermarking is suitable for ownership protection because the algorithm is robust against possible attacks. Fragile watermarking is suitable for detecting tampering [17]. The secure communication of bank documents and protection against tampering is necessary for banking applications. Zarrabi et al. [18] proposed blind and diagnostically lossless watermarking (BlessMark), and Fares et al. [19] proposed a blind and robust watermarking algorithm based on the DCT and Schur decomposition. Agarwal and Singh [20] presented a watermarking method by combining the DCT and genetic algorithms for optimization. The experimental results showed an increase of 12% to 18% in PSNR values. Singh [21] proposed a watermarking algorithm by combining the Discrete Wavelet Transform (DWT), Bi-dimensional Empirical-mode Decomposition (BEMD), Discrete Cosine Transform (DCT), Particle Swarm Optimization (PSO), and Singular Value Decomposition (SVD). The experimental results yielded high-quality watermarked images and good robustness against various attacks.

III. ADD-KYC: Automated deep decentralized know-your-customer

The proposed ADD-KYC framework consists of four main modules:

1. A decentralized blockchain-based module is used for sharing KYC information and machine learning models in the interbank ecosystem among various parties. This module protects KYC data by providing blockchain-based encryption and a transfer system that allows the customers to have full control over their KYC data.
2. A deep neural network-based biometrics validation module is used to authenticate customers and financial documents, such as bank checks.

TABLE 2. Comparison of ADD-KYC with existing blockchain-based KYC frameworks.

Reference	Automated KYC	Multi-biometrics-based KYC	Hierarchical anonymisation of	IoRT-based KYC collection	KYC on blockchain
ADD-KYC	✓	✓	✓	✓	✓
Biradar et al. [13]	×	×	×	×	✓
Mamun et al. [22]	×	×	×	×	✓
Kapsoulis et al. [23]	×	×	×	×	✓
Yadav et al. [14]	×	×	×	×	✓
Geroge et al. [24]	×	×	×	×	✓

3. A high-capacity lattice vector quantization-based watermarking module is used to enable the secure transmission and archiving of various banking documents.
4. A privacy-preserving module is used to anonymise people in videos collected by IoRT agents in the banking environment for privacy-aware hierarchical access.

These modules are detailed below.

A. Decentralized and secure KYC communication module

From the perspective of information technology, creating trust among multiple parties without involving a third party has been a challenge for many years. The emergence of crypto currencies has led to the introduction of the blockchain as a solution for decentralized environments. Smart contracts have expanded the use of the blockchain by running a code in a decentralized manner and making all the parties reach a consensus on the results of its execution. In smart contract systems like Ethereum, the execution of every code instruction is verified by the members of the blockchain in a decentralized network and incentivized by the currency of the network. This combination of decentralized execution and incentivization has led to the development of various decentralized applications.

The smart contract is a transaction protocol that executes automatically and performs various operations, like funds transfer, calculations, and information storage [25]. It applies to multiple fields, including smart homes, asset management, and e-commerce [26]. The smart contract includes the value, address, functions, and state of transactions executed between participating nodes. It accepts transactions as input, applies a function to execute the corresponding code, and provides the output. The smart contract is a successful application of blockchain technology that reduces the cost of transactions and execution time and increases security [26].

The proposed ADD-KYC uses a combination of blockchain and deep neural networks to increase the speed and accuracy

of KYC. When a customer opens an account in the bank, the process of KYC begins. It requires an identification card and proof of address as well as other documents depending on the level of KYC. To prove their identity, the customer provides documents like a passport or driver's license, and the proof of address is usually provided in the form of a recent bill. The process can be carried out online by uploading the documents to the bank's website or offline at the branch. In ADD-KYC, a series of biometrics are collected from the customer in addition to these documents. A photograph of the customer for facial recognition, a sample of their fingerprints and voice, and a signature is collected. However, other biometrics, such as the iris, finger veins, and palm veins, can be added to the ADD-KYC in future versions to increase the accuracy of the system. The collection and verification of intrusive biometrics usually reduce customer satisfaction with the KYC procedure and increase the likelihood of the bank losing the customer. After the initial phase of customer KYC registration, a pair of public and private keys are assigned to the customer.

All entities participating in the ADD-KYC use asymmetric cryptography for security on the blockchain. Each entity in asymmetric cryptography has a pair of keys: the public key, which acts as the address of the entity, and the private key, which is known only to the owner and is used for securing the data. The ADD-KYC uses the Stellar blockchain network, the standard for cryptography which is a version of the Edwards-curve Digital Signature Algorithm (EdDSA), called Ed25519. It uses a combination of SHA-512 and Curve25519. Using this key, the bank issues a traditional KYC token containing the customer's validated identification card and proof of address. The KYC token is encrypted with the customer's public key and signed with the bank's private key. This traditional KYC token is used by the customer for initial authorization of the KYC procedure in interbank financial services. For simplicity of service and comfort of the customer, it is assumed that the KYC token is not required for the provision of intrabank services. The biometrics data collected in the registration stage are used as a mini-batch for the online training of the biometric deep neural networks of the bank.

As shown in FIGURE 1, the Interplanetary File System (IPFS) [27] is used in ADD-KYC for the transmission of biometric records, customer banking records, and trained models. For higher security, the bank documents are watermarked with the customer's KYC information using the proposed high-capacity lattice vector quantization algorithm. The IPFS is a protocol designed for decentralized file sharing that does not have the problems of current-centralized (client-server) HTTP architectures. It is a decentralized peer-to-peer protocol that disseminates files among a list of trusted nodes, known as bootstrap nodes and makes them available to other users through Content Identifiers (CIs) and content-based addressing.

As mentioned above, each bank's mainframe that trains, stores, and validates the deep biometric models has a public and a private key. Every IoRT agent and each customer has

public and private keys. It is assumed that the customer uses the bank that has the customer biometrics model (the customer and robot belong to Bank-A) for the provision of regular services. The scenario for interbank biometric validation is discussed later.

Once the customer of Bank-A has been approached by the bank's service robot, biometrics of their face, voice, signature, and fingerprints are recorded unobtrusively by it. The robot encrypts the biometric data with the public key of Bank-A and sends the results to the IPFS. The IPFS returns a hash (address) and the robot sends the hash to the mainframe of Bank-A. The mainframe uses this hash to receive biometric data from the IPFS and decrypts it with the private key of Bank-A. The mainframe validates the identity of the customer using the deep biometric model and sends the customer's identity and authorization for the provision of financial services, encrypted with the robot's public key, over the IPFS to the robot. The robot decrypts the customer's identification and authorization number using its private key and greets the customer with the provided name and bank ID to start offering services to them. Asymmetric cipher algorithms are not suitable for the encryption of large biometric files and symmetric ciphers are used. In the proposed framework, the AES [28] algorithm is used for symmetric encryption. In symmetric algorithms, a common key is used on both sides. The symmetric key is composed of an eight-digit random number combined with a timestamp to create a unique encryption key for a large biometric file. The file is encrypted using the symmetric key and sent over to the IPFS. The IPFS returns the address of the file as a hash. A JSON file is created using a combination of this hash and the encryption key. Since any data stored in IPFS is publicly accessible, to preserve the confidentiality of the file, Rivest-Shamir-Adleman (RSA) asymmetric cipher is used. In asymmetric ciphers, every side of the transmission has a pair of public and private keys. Anything encrypted using the public key of the destination can only be decrypted by the private key of the destination. Therefore, the confidentiality of the data is preserved. Since the JSON file is encrypted using the public key of the destination bank, it can only be decrypted using the private key of that bank. For simplicity of the description of the ADD-KYC, from here on, this two-step symmetric-asymmetric encryption-decryption process is simply referred to as the encryption-decryption of files by public and private keys.

The issue of automated biometric-based interbank KYC validation can also be addressed using ADD-KYC. The main issue here is that the banks are legally not allowed to share the biometric information of their customers. However, there is no legal restriction on the sharing of deep biometric models between the mainframes of banks. In ADD-KYC, trained deep neural network models are shared between the mainframes of banks in a secure and decentralized framework. The IPFS is used to store checkpoints of the learned model.

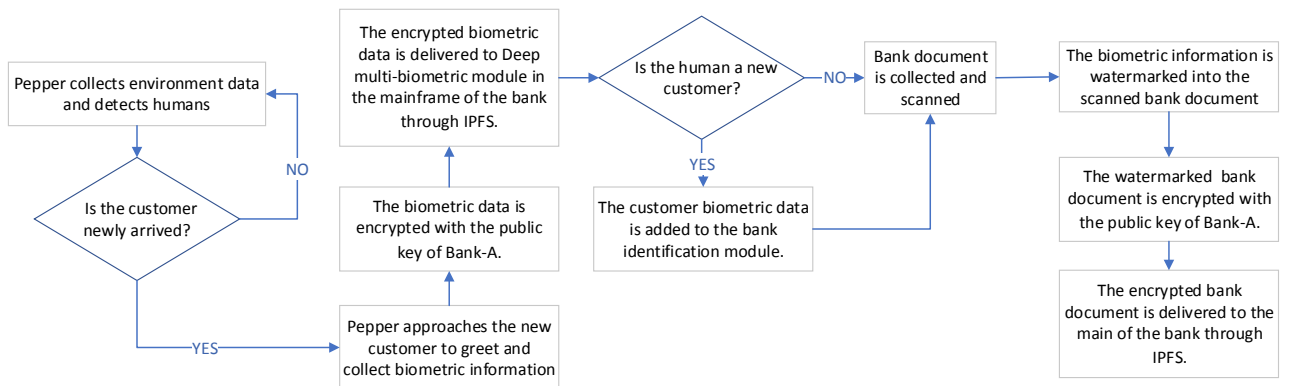


FIGURE 1. The block diagram of the proposed ADD-KYC framework for internal banking.

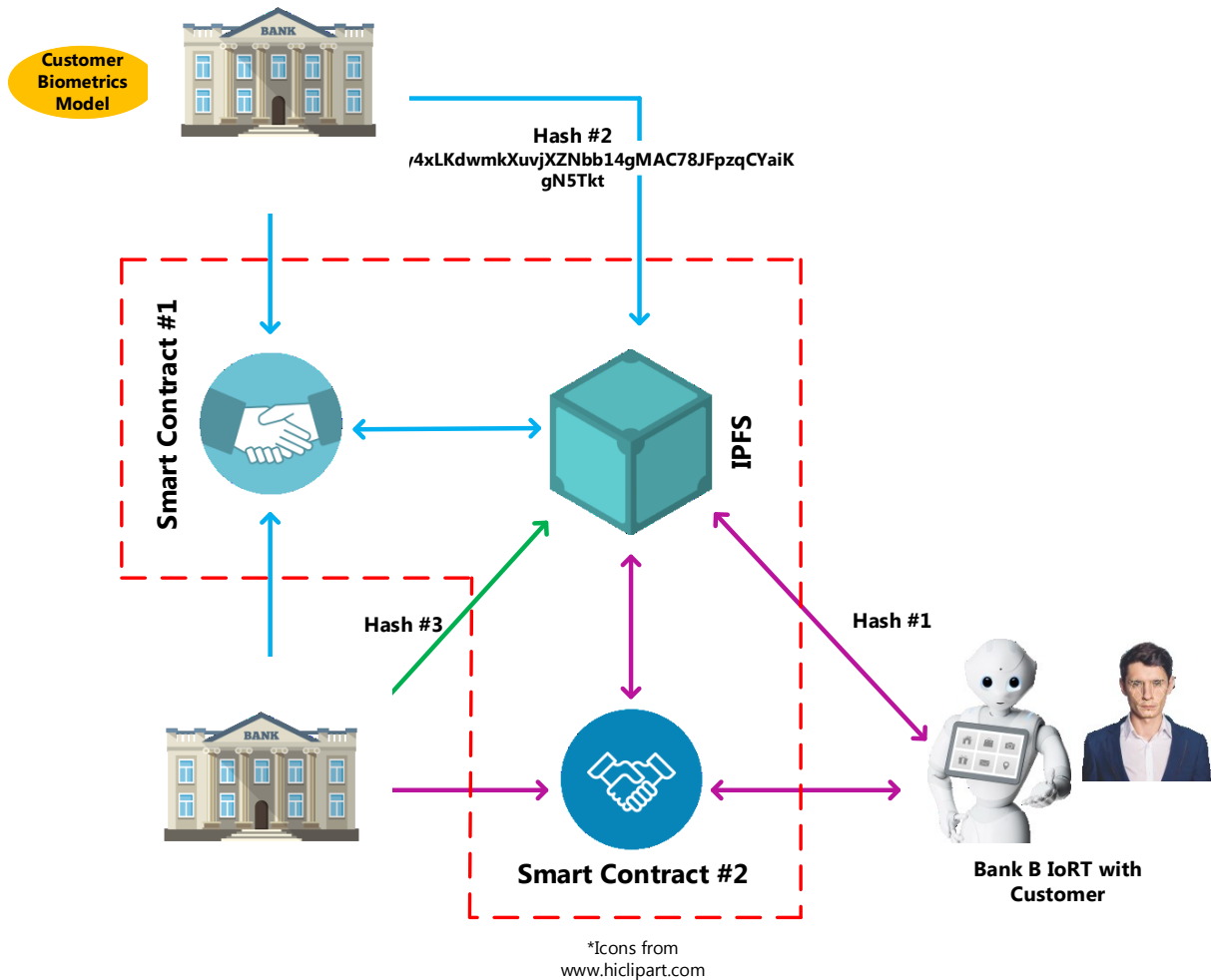


FIGURE 2. Interbank KYC based on ADD-KYC.

When the client whose biometric models have been initialized in Bank-A requests services from IoRT agents of Bank-B, the process starts with the provision of their public key to Bank-B. Bank-B sends the public key to the customer, and the customer's digital wallet encrypts their KYC token using this key and sends it to Bank-B over the IPFS. Bank-B decrypts the token using its private key and obtains the KYC of the customer. Bank-B then requests and receives the deep biometric model of the customer from Bank-A.

The trained checkpoints containing the model for the biometrics of the customer are transmitted from Bank-A to Bank-B over the IPFS and are validated using the public and private keys of both banks. This is similar to the decentralized transmission of biometric data and customer ID between the robot and the bank. The rest of the KYC is similar to the intrabank KYC. Figure 2 shows the framework for interbank KYC based on ADD-KYC.

B. The deep biometric verification module

The architecture used in this case study for the multi-biometric KYC module is a deep neural network for the feature extraction and integration of biometrics, as shown in FIGURE 3. A group of convolutional feature extractors is used in the first section of the architecture. Four modules are used in this study for modeling the customer's face, voice, fingerprint, and signature. All convolution layers apply 1D convolutions over time, and each has 32 kernels of size five. Moreover, max-

pooling with a size of five and a stride of two is used in all pooling layers. Finally, a Softmax layer is used to predict the probabilities of the output class. Each residual block contains two convolutional layers, two Rectified Linear Unit (ReLU) layers, a residual skip connection, and a pooling layer. These layers are in charge of extracting features from the biometric data. The integration layer consists of a dense layer and a Softmax layer.

Details of the collection of biometric data using Softbank's humanoid robot Pepper as well as the processing and extraction of visual, audio, and signature-related biometrics are presented in Sections IV-A and IV-B, respectively.

C. Lattice vector quantization-based high-capacity watermarking

The root lattice A_6 is a subset of points in 7D space such that the sum of their integer coordinates is zero, given as

$$A_6 = \left\{ (x_0, x_1, x_2, x_3, x_4, x_5, x_6) \in \mathbb{Z}^7 : \sum_{i=0}^6 x_i = 0 \right\} \quad (1)$$

The lattice points are determined by the generating matrix. There are two types of generating matrices, square ($n \times n$) and rectangular generators ($n \times n + 1$):

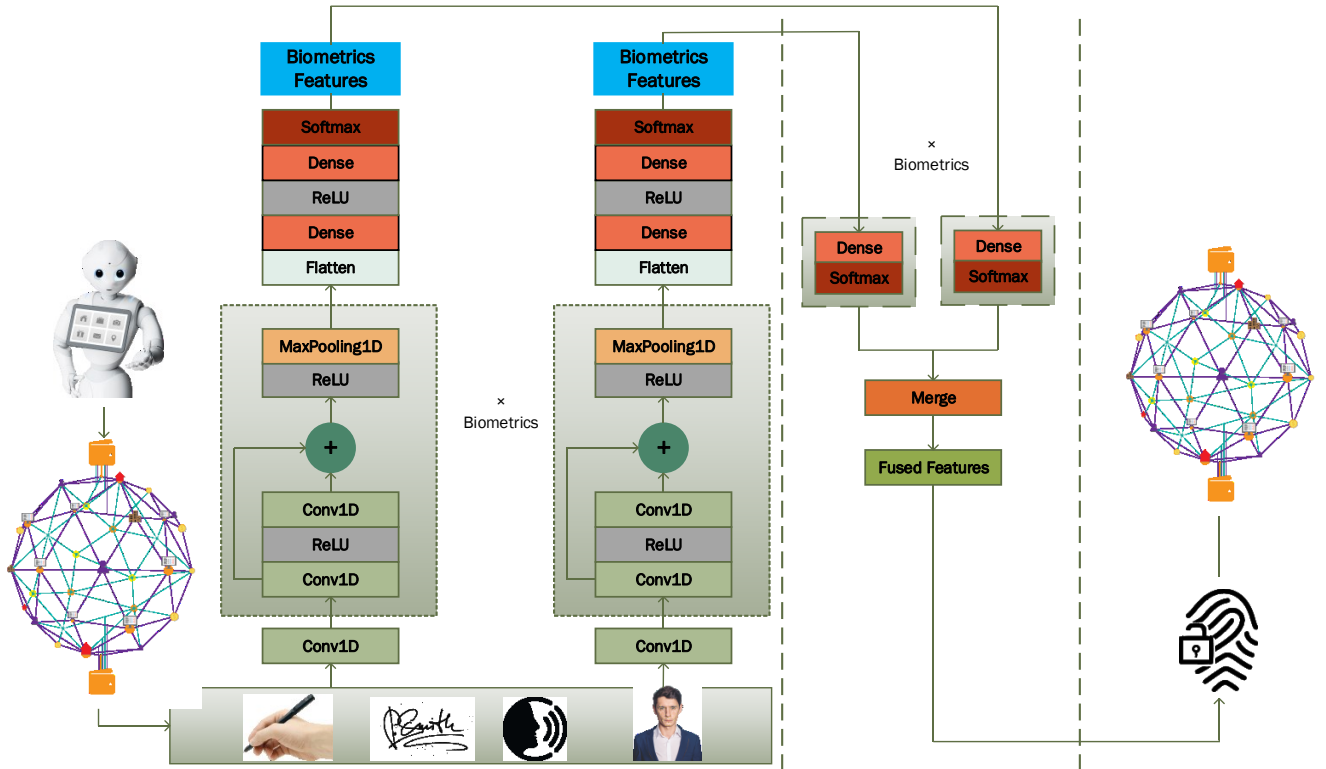


FIGURE 3. The architecture of the deep multi-biometric identity module.

$$\mathbf{G}_{A_{6,6 \times 7}} = \begin{pmatrix} -1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 1 \end{pmatrix} \quad (2)$$

$$\mathbf{G}_{A_{6,6 \times 6}} = \begin{pmatrix} 1.414 & 0 & 0 & 0 & 0 & 0 \\ -0.707 & 1.224 & 0 & 0 & 0 & 0 \\ 0 & -0.816 & 1.154 & 0 & 0 & 0 \\ 0 & 0 & -0.866 & 1.118 & 0 & 0 \\ 0 & 0 & 0 & -0.894 & 1.095 & 0 \\ 0 & 0 & 0 & 0 & -0.912 & 1.08 \end{pmatrix} \quad (3)$$

The Gramm matrix is defined as the multiplication of the generator by its transpose. $\mathbf{G}_{A_{6,6 \times 7}}$ and $\mathbf{G}_{A_{6,6 \times 6}}$ generate equivalent lattices because they have proportionate Gramm matrices. The sub-lattices of a root lattice are a subset of the original lattice but with the same symmetry. The ratio of the determinant of the sub-lattice generator to the determinant of the lattice generator is defined as the index N . For example, similar sub-lattices of A_6 with index $N = 64$ can be generated by

$$\mathbf{G}'_{A_6} = \begin{pmatrix} -0.707 & -1.224 & -0.866 & -1.565 & -1.278 & -1.080 \\ -0.707 & 0.408 & -1.443 & 1.565 & 1.278 & 1.080 \\ 0 & -1.633 & 2.309 & 0 & 0 & 0 \\ -0.707 & 2.041 & -0.288 & 0.670 & -1.278 & -1.080 \\ 0 & 0 & 0 & -1.788 & 2.190 & 0 \\ 0 & 0 & 0 & 0 & -1.825 & 2.160 \end{pmatrix} \quad (4)$$

To quantize the input points onto a specific lattice using fast quantization algorithms [29], it is necessary to find their corresponding transformation matrix. The transformation matrix projects the n -dimensional input vector onto $n+1$ -dimensional vectors on the $\sum_{i=1}^{n+1} x_i = 0, x_i \in \mathbb{R}$ hyper-plane using matrix-multiplication [29]. The transformation matrix (T_i) can be obtained by:

$$\mathbf{T} = \text{inv}(\mathbf{G}_{A_{6,6 \times 6}}) \times \mathbf{G}_{A_{6,6 \times 7}} = \begin{pmatrix} 0 & 0 & 0 & 0.500 & 0.500 & -0.500 & -0.500 \\ 0 & 0.577 & 0.577 & -0.288 & 0.288 & -0.288 & -0.288 \\ -0.612 & -0.204 & 0.408 & -0.204 & 0.408 & -0.204 & 0.408 \\ -0.474 & -0.158 & 0.316 & 0.474 & -0.316 & 0.474 & -0.316 \\ -0.387 & 0.516 & -0.387 & -0.258 & 0.387 & 0.387 & -0.258 \\ -0.327 & 0.436 & -0.327 & 0.436 & -0.327 & -0.327 & 0.436 \end{pmatrix} \quad (5)$$

The transformation matrix for a similar sub-lattice can also be determined by the same equation using the corresponding generator.

We now describe the embedding procedure based on the Integer-Lifting DWT (ILDWT) and LVQ. The RGB cover image—in this case, the bank document—is first digitized and resized to 1093×2372 pixels. Then, the red, green, and blue planes are processed separately. Each plane is transformed by the ILDWT to form four sub-bands, namely, CA, CH, CV, and CD. Each sub-band has dimensions of 544×1186 . Because the embedding process is defined for 6D vectors, we need to have 6×6 blocks. Thus, every sub-band is reshaped to 17,640 square blocks and then quantized by the LVQ algorithm onto a sub-lattice with an index of 256 or 4,096.

Assume that the first row of the first block is V_i . The corresponding quantized vector is $V_{i,lvq}$. The Voronoi of every sub-lattice point includes many Z_6 points that are all quantized on the same sub-lattice point $V_{i,lvq}$. In other words, these points can be interchanged without any difference in the quantization process. Thus, if the set of Z_6 points inside the Voronoi of a sub-lattice point are found, it is possible to replace them with another member of the same set according to the payload data using an appropriate mapping scheme, which is as follows: The set of all Z_6 points around the origin are found and sorted according to their distance from it. This set is saved as the mapping. To find the set of Z_6 points within the Voronoi of a sub-lattice point, $V_{i,lvq}$ is added to every row of the mapping table and quantized. If it is also quantized on $V_{i,lvq}$, then the corresponding row is added to the set of Z_6 points. All points in the set of Z_6 points are very close to one another and, consequently, impose low distortion on the cover image. If, for example, the payload data is 112, the 112th member of the set of Z_6 points replace the input vector V_i .

For watermarking bank checks, the payload data are composed of several items. The first is the fingerprint of the customer. Each fingerprint image is a grayscale image with dimensions of 800×750 pixels and adds 600,000 values to the payload array. The second item is the name of the recipient of the check and adds 128 values to the payload. The third item in the payload is the check number, which adds four values to the payload. The fourth part is the signature of the customer. Signature images have a size of 286×284 pixels. These images are resized to 128×128 , and add 16,384 values to the payload. The fifth part of the payload includes 187,500 values that belong to the facial image of the customer. The last part of the payload includes the logo of the bank. It contributes 94,248 values to the payload. The embedding process is carried out on all blocks of the three planes. However, the CA sub-bands are not used because they impose large distortions. In the end, wavelet sub-bands are inversely transformed, and the cover image of the check with the embedded data is formed.

The same procedure as the above is performed to extract the embedded data. The cover image of the check is first transformed by ILDWT. Then, the wavelet sub-bands CH, CV, and CD are reshaped to form 6×6 blocks. Every row of blocks is quantized, and the set of all Z_6 points that reside in the Voronoi of the quantized vector are formed. Finally, the input vector is searched for in the set of all Z_6 points. If the vector is in the m^{th} row, m data are extracted.

IV. Deep, hierarchal, visual privacy-preserving module

The IoRT banking robots collect visual biometrics while monitoring the banking environment. This process creates large video databases, because of which the issue of secure transmission and storage as well as access to these data needs to be addressed in the sharing layer of the IoT for smart banking. Various departments, including security and business intelligence, use these video data in the banking sector.

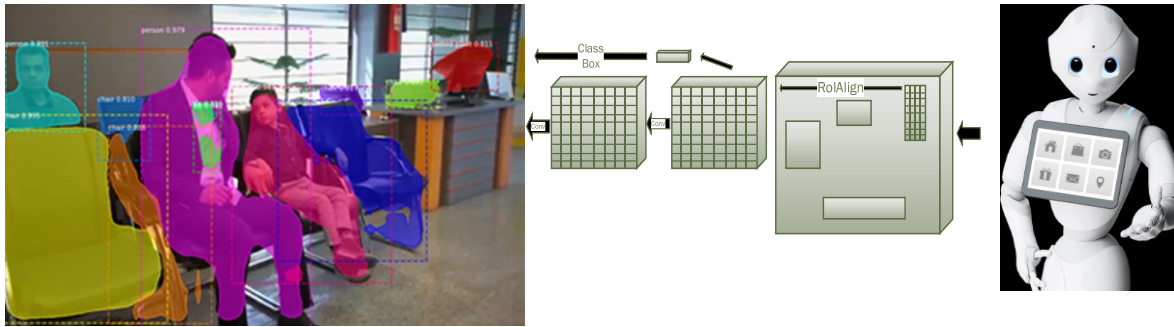


FIGURE 4. Deep privacy preservation for the IoRT in the environment of the bank branch.

The collection of these videos from inside the bank branches creates significant privacy issues. Various solutions for privacy preservation in the IoT have been presented [30].

To preserve the privacy of bank customers appearing in the recorded videos, a deep Mask R-CNN [31] is used in the proposed framework for segmenting and labeling various objects in these videos [32]. The permission level granted to each department of the bank is then defined. At the highest level are the legal authorities that can see the identity of all individuals in the videos. The bank's security department also has access to all levels of data. The business intelligence department and other departments that have signed Non-Disclosure Agreements (NDA) contracts have a medium level of access. Ordinary staff and other departments have access to only fully masked datasets. FIGURE 4 shows the visual privacy protection module for the IoRT in the environment of the bank branch.

V. Experimental setup and results

A. Experimental setup

• Automated Bank Branch Simulation Environment

We tested the application of humanoid service robots for smart banking in collaboration with the Bank Pasargad of Iran for over one year. As shown in FIGURE 5, an environment similar to that in a branch of Bank Pasargad was created to simulate the interactions between customers and the robot. The humanoid robot Pepper from Softbank Robotics was used as the service robot for KYC verification and the provision of financial services to customers.

The Pepper robot is used for greeting customers and collecting biometric data and bank documents. The privacy-preserving module, the multi-biometric customer identification module, the secure symmetric/asymmetric biometrics and document transfer module over IPFS, and LVQ based watermarking module, which are the components of the proposed ADD-KYC framework, required processing power higher than the capabilities of the pepper robot and are executed on banks computer systems.

• Datasets

Traditional clothing dictates partial coverage of the heads in Middle Eastern countries. Headscarves and similar items

partially cover the head and part of the person's face. Therefore, one of the major issues for person identification and re-identification in the Iranian bank environment is the challenges posed by the dress codes. To address this issue, a dataset for person identification in indoor environments consisting of 5,000 images is collected.

To provide reproducible results and also due to privacy concerns and limitations, in this research for bank check datasets and fingerprint datasets, a series of publicly available datasets are used for watermarking simulations and experimental results. The publicly available IDRBT Cheque Image Dataset is used in this research as the check dataset [33]. The images of checks in the dataset were TIFF files with dimensions of $1093 \times 2372 \times 3$ pixels. A total of 112 checks leaves from four different banks in India are used as a source document. The dataset is completely simulated to avoid any privacy concerns. The dataset is only used for presenting the watermarking results and the geographical location of the dataset does not affect the watermarking procedures. The checks are cover images for watermarking and the embedding information is digital biometrics data. The Tsinghua Distorted Fingerprint Database was used as the fingerprint dataset [34]. The dataset contains a total of 320 pairs of normal fingerprints and distorted fingerprints from 185 different fingers. For Persian handwriting, signature and check values recognition a series of actual Iranian checks are used in this research. The images of these checks are not included in this manuscript due to privacy concerns.

B. Facial and vocal biometrics extraction using a humanoid robot

Pepper was controlled using the Aldebaran Choregraphe Suite [35]. The out-of-box robot is not capable of any procedure required for biometric collection adhering to the banking requirements and extensive programming is required for adding these capabilities. The robot is programmed to be able to perform basic maneuvers in the simulated bank branch, greet the bank's customers, and answer frequently asked banking questions. Autonomous question-answering by the robot is currently being explored as a future stage of its interaction with customers [36].

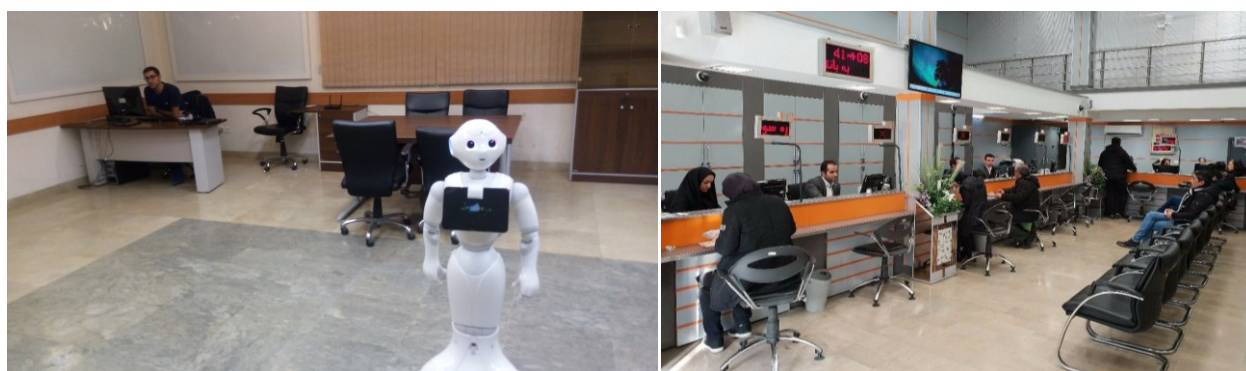


FIGURE 5. Simulated environment for the banking service robot compared with the environment in an actual bank branch.

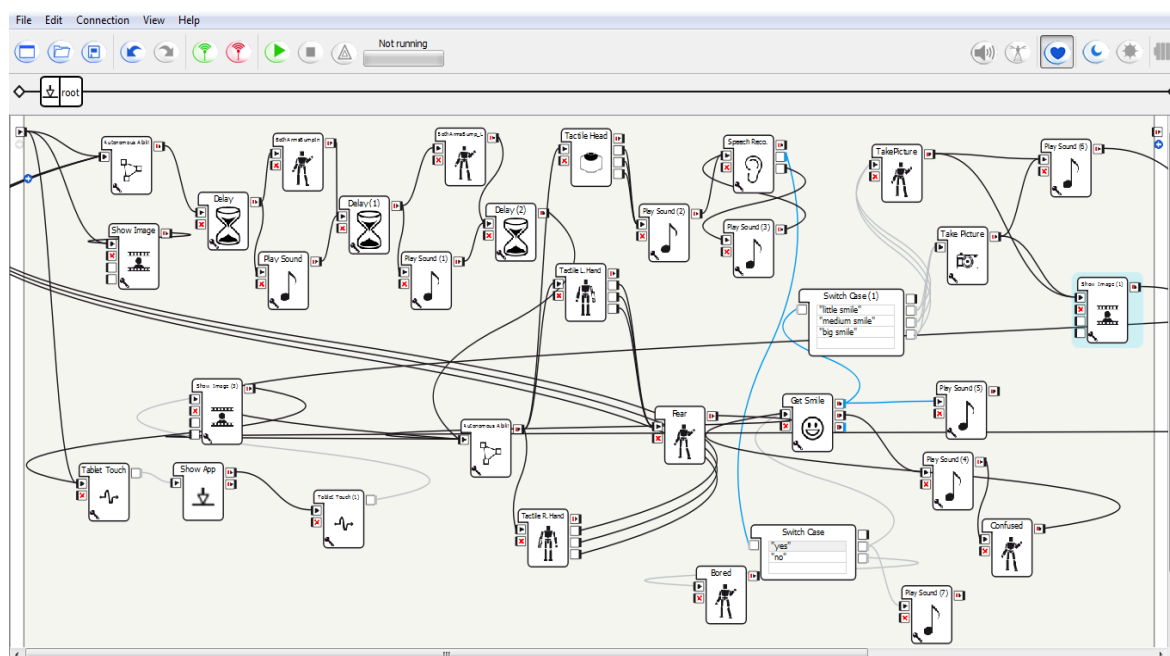


FIGURE 6. State flow for the robot's maneuvers, greeting, and other basic operations.

The design of state flow for these operations in the Choregraphe Suite environment is presented in FIGURE 6. Pepper has a face detection and a face recognition module in the Choregraphe Suite. Face detection works automatically. When a face is detected, an ID value is assigned to it. This Choregraphe Suite was used to collect facial and vocal biometrics as shown in FIGURE 7.

Facial data in the video were used to generate samples for training the initial biometric model of the bank's customer. To prevent the deep neural model from overfitting and increase the amount of data in our dataset, we used data augmentation techniques, including random shearing, random rotation, and random translation. Keras and TensorFlow were used for the deep biometrics module. To improve the accuracy of facial

recognition, a combination of the Haar cascade and the Locally Binary Pattern Histogram (LBPH) for feature extraction led to acceptable results [37]. The detection of the facial emotions of the customers is also under investigation as a future stage of the project [38].

C. Watermarking of bank documents with results of customer information

We evaluated the performance of the proposed watermarking algorithm and compared the results with several recent algorithms proposed in the literature. To be able to compare their performance, we used six standard grayscale images: Lena, Baboon, Airplane, Elaine, Lake, and Boat. In the

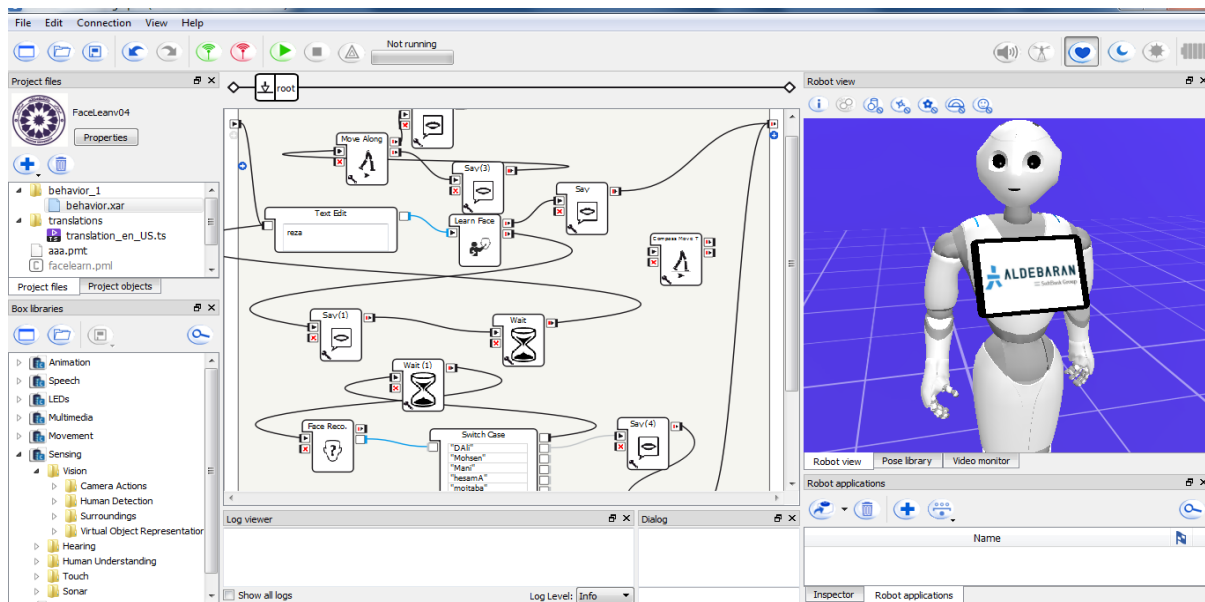


FIGURE 7. The state flow of Choregraphe Suite for the collection of facial and vocal biometrics.

experiment, only horizontal sub-bands were used for embedding.

FIGURE 8 compares the results of our method with those of algorithms proposed in Refs. [39], [40], [40], and [17]. All images used had a size of 512×512 and were downloaded from the USC-SIPI database [41]. We started with a low payload of data by embedding them in horizontal sub-bands of the first block and continued by increasing the number of embedded blocks to 32, 64, 96, 128, 256, 384, 512, 640, 768, 896, 1,024, 1,280, 1,536, and 1,764. FIGURE 8 shows that the proposed method, with indices $N=256$ and $N=4096$, outperformed the other four algorithms. The performance of the method proposed in [17] was affected by the quantization error as well as the embedding process. The 6D LVQ was more spherical than the 4D LVQ, which resulted in a lower distortion in it. The proposed embedding process produced much lower distortion compared with the method in [17]. Therefore, the total imposed distortion did not generate overflow or underflow. This enhanced the performance of the proposed

algorithm in terms of the PSNR between the original images and the embedded images. For example, the proposed algorithm had values of PSNR that were 2.72 dB and 1.9 dB higher on the Baboon and Lena images, respectively, than the method in [17] for the same amount of embedded data. Details of comparisons with the other methods based on a high embedding rate are presented in Table 3.

Table 3 shows that the proposed algorithm, with $N = 1048576$, had a capacity that was higher by 0.65 bpp on average than that of the method proposed by Govind et al. [42], with a PSNR that was lower by 4.9 dB. However, with $N = 32768$, the results were slightly better, with the proposed method recording an embedding capacity that was 0.06 bpp higher and a PSNR that was 1.5 dB higher. Table 4 presents the comparative results based on a low embedding rate.

TABLE 3. Comparisons based on a high embedding rate.

Scheme		Lena	Baboon	Boat	Elaine
Lee et al. [43]	Capacity (bpp)	1.4703	2.438	1.4664	1.590
	PSNR (db)	29.940	21.710	27.7489	30.086
Zhang et al. [44]	Capacity (bpp)	1.7576	1.9023	1.7312	1.370
	PSNR (db)	31.436	21.220	27.4470	31.023
Shaik et al. [45]	Capacity (bpp)	1.8368	2.4124	1.8944	1.896
	PSNR (db)	32.244	26.654	28.5130	30.640
Wahed et al. [46]	Capacity (bpp)	1.751	2.327	1.939	1.873
	PSNR (db)	35.41	29.741	34.60	35.05
Govind et al. [42]	Capacity (bpp)	2.301	2.454	2.2982	2.2854
	PSNR (db)	38.747	30.104	38.6965	38.712
Proposed N = 256	Capacity (bpp)	0.969	0.97	0.969	0.969
	PSNR (db)	46.69	46.53	46.61	46.58
Proposed N = 4096	Capacity (bpp)	1.090	1.09	1.09	1.09
	PSNR (db)	44.89	44.80	44.83	44.83

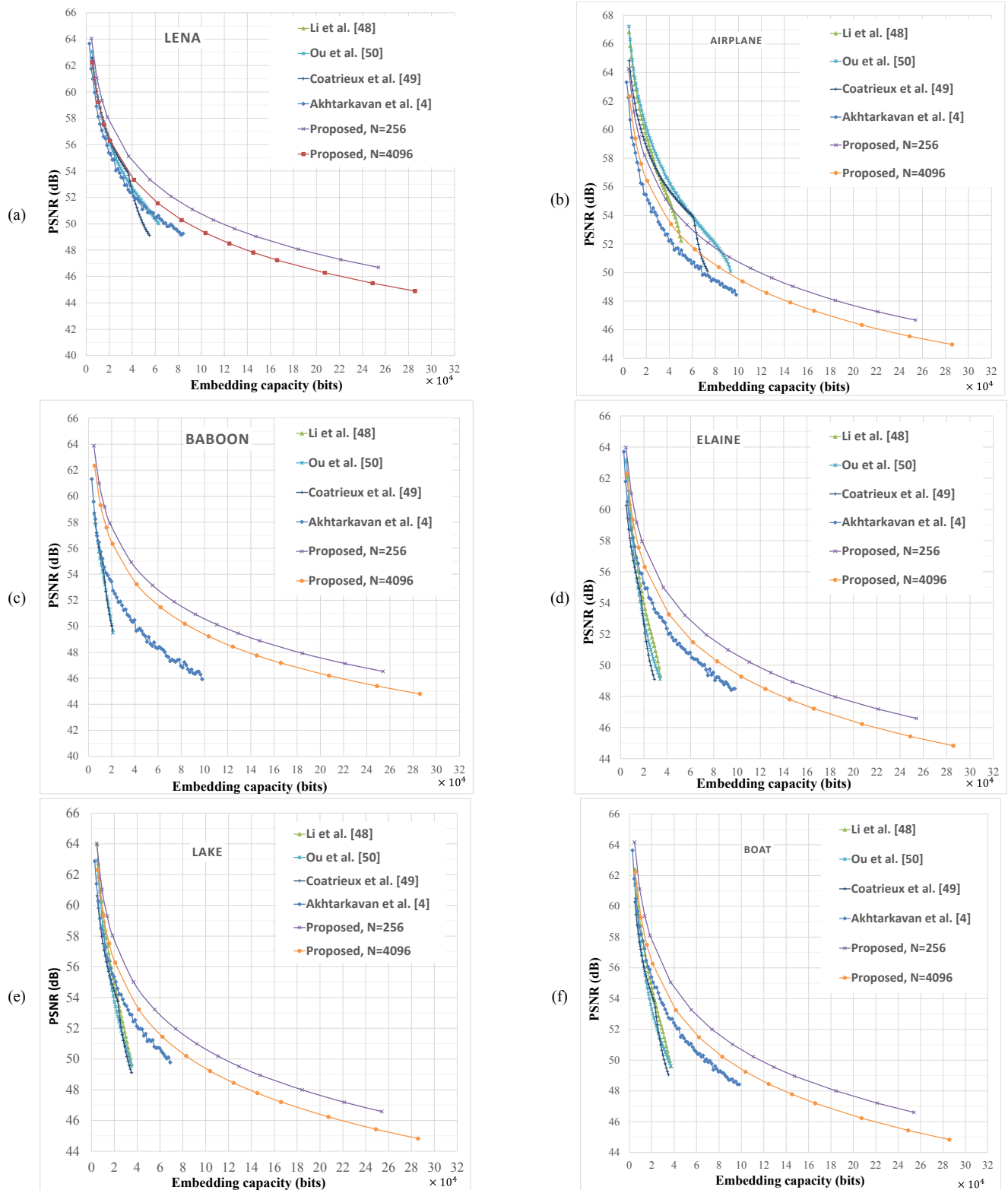


FIGURE 8. Comparison of the performance of the proposed algorithm with prevalent methods on (a) Lena, (b) Airplane, (c) Baboon, (d) Elaine, (e) Lake, and (f) Boat.

Table 4 shows that the proposed algorithm with $N = 256$ delivered better performance than the method by Govind et al.

[47], with an embedding capacity that was 1 BPP lower on average but a PSNR that was 5.1 dB higher. However, with N

= 4096, its embedding capacity was on average 0.8 lower and the PSNR was 3.5 dB higher.

In the watermarking module of the proposed framework, information about the customer, including their fingerprint and name, is embedded as a watermark in the related bank documents, such as checks. It is transmitted using the IoRT-based BaaS and archived in the bank's digital database. This procedure protects the data against tampering and other forms of security attacks. To ensure that the privacy of customers was protected, we conducted experiments on a publicly available dataset of bank checks [33]. The images of checks in the dataset were TIFF files with dimensions of $1093 \times 2372 \times 3$ pixels. In addition, a publicly available dataset of fingerprints was used [34]. The fingerprints were BMP files with dimensions of 800×750 pixels. To quantify the extent of distortion between the original checks and their images with the embedded data, the PSNR was calculated and presented in Table 5. A higher value of PSNR indicates lower distortion. The results for images of 100 checks yielded an average PSNR of 45.5 dB after embedding, which indicated low distortion in the images after watermarking. Embedding the customers' information in 20 checks with two embedding algorithms, with indices of 256 and 4,096, led to PSNR values of 40.55 and 40.24 dB, respectively.

The simulations were performed on a computer system with an Intel(R) Core (TM) i9-10850K CPU @ 3.60 GHz, with 64 GB of RAM, Windows 10, and MATLAB R2019b. The average time for the watermarking procedure was 89.99 ms.

D. Autonomous check processing and verification using ADD-KYC

Despite the increasing use of the Internet and electronic banking, handwritten checks are still used in many financial and commercial transactions. These are usually either transactions in which a financial transfer occurs after the signing of a contract or those in which a payment is transmitted later than the actual date on which the contract was signed. An important characteristic of check-based transactions is that they are usually large transactions and, therefore, their security is important. In the initial months of 2020, despite the prevalence of COVID-19 and the reduction in trade by March of the year, a total of 91.1 million interbank checks worth 1156 trillion Rials were processed in Iran. Even though they provided online banking services, many banks were forced to keep their branches open and provide face-to-face services for customers. One of the main concerns during this time was related to the processing of handwritten bank checks.

The problem of KYC for the automated processing of bank checks consists of two parts. First, the customer who brings the check to the bank should pass the KYC. Second, the check needs to be verified to prevent forgery and fraud. Usually, the biometric KYC of the check is interbank. After verification of the customer with the bank check by the robot using the methods described in Section 3, the check is scanned and the process of verification starts. The first piece of information that needs to be verified is the name of the bank issuing the check. As shown in FIGURE 9, this process can be performed using a simple matching of the logo of the check with those of Iranian banks.

TABLE 4. Comparisons based on a low embedding rate.

Scheme		Lena	Baboon	Boat	Elaine
Chuen et al. [39]	Capacity (bpp)	0.952	0.709	0.868	0.871
	PSNR (db)	33.92	28.39	30.37	30.10
Zhang et al. [44]	Capacity (bpp)	0.878	1.784	0.803	0.890
	PSNR (db)	33.395	22.021	29.029	31.888
Shaik et al. [45]	Capacity (bpp)	1.514	1.221	1.218	1.307
	PSNR (db)	31.610	32.673	29.045	31.204
Chang et al. [48]	Capacity (bpp)	0.942	1.513	1.027	0.937
	PSNR (db)	43.61	34.63	41.12	42.79
Wahed et al. [46]	Capacity (bpp)	1.245	1.585	1.360	1.215
	PSNR (db)	41.38	34.75	40.50	41.02
Govind et al. [42]	Capacity (bpp)	1.571	1.667	1.562	1.539
	PSNR (db)	43.861	44.910	43.801	43.704
Proposed N = 256	Capacity (bpp)	0.563	0.56	0.563	0.563
	PSNR (db)	49.04	48.89	48.96	48.93
Proposed N = 4096	Capacity (bpp)	0.633	0.63	0.63	0.63
	PSNR (db)	47.24	47.18	47.20	47.21

TABLE 5. Original images of checks, fingerprints, and names, the embedded images of the checks, and the PSNR values.

Original check	Fingerprint	Customer info	PSNR
		Pradeep Kumar	45.62 dB
		Jaya Krishna	45.61 dB
		Prasanna Kumar	45.81 dB
		Kumar Ravi	45.78 dB
		Manish Ojha	45.76 dB



FIGURE 9. Matching the logo of the check with one of the logos of Iranian banks.

In interbank KYC for check verification, Bank-A is the bank that receives the check and requests verification for it from Bank-B, which is the issuer of the check. First, the customer uses the traditional KYC token to initialize the ADD-KYC procedure. Then, Bank-A encrypts the image of the check using the public key of Bank-B and sends it over to the IPFS.

The resulting hash is sent to Bank-B by Bank-A and used to access the image of the check. This image is decrypted using the private key of Bank-B, and the process of the analysis and verification of the check is started on the mainframe of Bank-B. After verification of the signature of the issuer of the check, the authorization for the deposition of the amount of the check

is sent to Bank-A by Bank-B using the IPFS and blockchain, as described in Section 3.

The challenges posed by the recognition and verification of handwritten characters and signatures are emphasized in handwritten bank checks, as many involve poor handwriting, are damaged because they are folded and put into wallets, and have other complications. The autonomous analysis of handwritten bank checks is thus a topic of research in the area. Mistry et al. [49] proposed using capsule networks for recognizing the amounts for which bank checks are written. Dhanawade et al. [50] used such image processing techniques as thinning, median filtering, and dilation to extract the amount for which checks are written. Srivastava et al. [51] used the 2D Convolutional Neural Network (CNN) for optical character recognition on checks. Jha et al. [52] used CNN to extract handwritten characters on checks. Holí et al. [53] proposed using CNN for identifying handwritten digits in bank checks. Wadhwa et al. [54] proposed using geometrical and structural features, like stroke width and direction, with the radial-basis SVM and random forest to verify the style of writing on bank checks. Neelima et al. [55] used the CNN for number recognition on checks, and Ghosh et al. [56] proposed a method for identifying handwritten characters on checks issued by banks in India using the support vector machine-based classification of texture through the histogram of oriented gradients and the grey level co-occurrence matrix. Bhat et al. [57] proposed an information integration-based

method to identify the amount for which bank checks are written.

We divide the autonomous analysis of checks into the following sections: 1) the autonomous analysis of handwritten checks, including the date, amount, machine-generated check number, and the name of the account holder; and 2) the validation of the handwriting of the bank's customer and signature. Handwritten parts of checks are extracted from scanned images in the first part of the check analysis module, as shown in FIGURE 10.

OpenCV v3.0 is used for processing the images of checks in this paper. The coordinates of bounding rectangles of the handwritten dates and the amount of the check are extracted as shown in FIGURE 11. We then use the deep neural network detailed in Section 3 and a Persian numbers dataset [58]. The proposed method can attain an accuracy of number recognition of 94%.

We then determine the amount of the check. Identifying the amount is similar to identifying a date, but is more challenging. The issues of non-compliance with writing numbers in check tables as well as poor handwriting pose the first challenge. Many people use different symbols, such as decimals and cross signs, as well as continuous lines instead of the number zero. The position of each number in the written amount has a significant impact on the output. Examples of non-standard amounts and amount recognition are provided in FIGURE 12.



FIGURE 10. The extracted handwritten parts of checks.

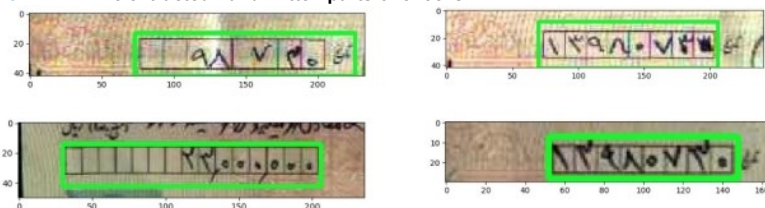


FIGURE 11. The extracted bounding rectangles of the handwritten dates and check amount.



FIGURE 12. Examples of non-standard amounts and amount recognition.

The next part of this module involves the interpretation of the machine-generated parts of the check. Because this part contains typed phrases, we used Tesseract optical character recognition to convert these phrases into text. The machine-generated portion of the check usually consists of two parts: The first includes the name of the account holder (name and surname), and the second part consists of the account number, which in some cases is in Persian and others are in English. These two sections are processed separately. The final stage involves signature recognition and verification. The main problem in signature detection is that users do not sign in at the place specified for this purpose. The signature usually overlaps with printed text on the check, gleaming the signature from it poses a challenge. A series of stages of pre-processing leads to signature extraction from images of the check. The outputs of signature identification and detection of the handwritten part are validated using the deep neural network presented in Section 3 to verify the issuer of the check. To prevent data tampering, the checks are watermarked with the customer's biometric information.

To evaluate the proposed, ADD-KYC in various banking scenarios, we performed a series of simulations. Its accuracy in verifying facial biometrics was acceptable at 93%, but that of vocal biometrics was poor at 78%. A combination of these biometrics resulted in an accuracy of 95%. Its accuracy of check verification was 82%. This low value was obtained due to the complexities of handwriting in Persian and signature recognition. The main limitation of the proposed solution is that in commercial banking, even one unsatisfied customer is too many, and therefore it is necessary to add a fail-safe mechanism to the ADD-KYC. A confidence level needs to be determined based on the probability of the correct verification of KYC. If this confidence level is lower than a given threshold, and there is the possibility of a customer being unsatisfied, the process should shift to manual service to that customer by one of the officials of the bank. The accuracy of the deep hierarchal privacy-preserving module was 94%, which is acceptable for commercial scenarios.

The cost of the robot and hardware used in this research is approximately less than 2500 USD. More advanced robots will cost around 15K USD. The average salary of a lowest-level bank clerk in the developed world is 30K+ USD per year. Therefore, similar to other human jobs replaced by robots, the cost-benefit of automation over time is significant. However, there can be a discussion of the social cost of automation which is out of the scope of this manuscript.

VI. CONCLUSION

The proposed framework was implemented using the humanoid robot Pepper, developed by Softbank Robotics, as a solution for cyber-physical banking during pandemic conditions. Owing to its lack of intrusiveness, it improves customer experience and efficiency while reducing costs in the context of physical banking. We evaluated the performance of the proposed watermarking algorithm and compared the

results with several recent algorithms proposed in the literature. For watermarking bank checks, the biometric security data is composed of the fingerprint of the customer, the name of the recipient of the check, the check number, the signature of the customer, the facial image of the customer, and the logo of the bank. The proposed algorithm provides PSNR values that were up to 5.1 dB higher than similar methods. The results for images of 100 checks yielded an average PSNR of 45.5 dB after embedding, which indicated low distortion in the images after watermarking. The results of the simulations showed that the proposed framework has an accuracy of more than 90%.

The main limitation of the proposed framework is that the banking standards require 100% satisfaction of the customers and therefore the automated system should always act under human supervision. Furthermore, the accuracy of the proposed framework is limited in its current state. Significant variations in customer biometrics and conditions of the bank environment require further training of models. The proposed solution at the current stage is proposed for banking service provision during the COVID-19 pandemic. In these conditions, providing regular KYC verification and paper-based banking services using human staff was a health hazard for both customers and the staff. In normal conditions determination of improvement in customer experience requires a large study on various banking environments and cultures. Large scale customer feedback collection and analysis is considered as further stages of this research and after implementation in several bank branches. Increasing the autonomy of interactions with the bank customers and inclusion of a greater variety of banking services are also under consideration as future works of this research.

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